Urban Waste Collection Optimization Using Pathfinding Algorithms

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***Abstract*—The Waste Management Unit (WMU) considers efficient waste collection as its primary target because it shortens collection durations while lowering expenses and environmental impacts. This project develops waste collection routes with minimum travel time between different centers by accounting for vehicle space and road structure and traffic situations. The research included ten major locations in Bengaluru and a disposal site for analysis. A review of routing efficiency involved implementing analysis using Dijkstra's and Bellman-Ford alongside Floyd-Warshall and Johnson's and Nearest Neighbour and Ant Colony Optimization (ACO). The metrics included travel time together with fuel usage and CO emission and overall route performance when evaluating these algorithms. This project contributes to wider objectives which merge optimization approaches with operational limitations to boost efficiency in fuel usage and emission reduction along with waste management systems in urban areas.**

***Index Terms*—Urban Waste, Shortest Path, Collection Routes, Bengaluru, Smart City**

I. Introduction

Operational challenges to waste management systems increase as cities expand their population and their spaces at the same time. Health issues from waste management operations require operators to implement medical protection measures into their current focus of efficiency and environ- mental sustainability. Two factors are omitted from current waste management scheduling systems despite operating along set routes according to established timetables. The factors are uneven waste generation levels alongside traffic congestion issues. The lack of careful waste system management causes dual sustainability and performance problems that increase operational time and fuel costs and service periods. Standard operations employed by current city waste collec- tion systems fail to measure up since they handle both chang- ing waste volumes and traffic patterns inadequately Rigid static systems operate poorly because they force waste collection to take longer and consume more fuel while costing the operation more money. There are barriers regarding the use of real-time data in both applications for The inability to alter delivery routes together with these weaknesses becomes significantly more important for organizations to address efficiently. The operational requirements need an efficient combination with environmental considerations environment aspects.

- The development of the system required an initial identification of the item regarding efficient waste collection route points. data extraction of geographical coordinates, identification The process developed item sets for system architecture which included determining vehicles' holding limits be- fore calculating routes based on distance. distance matrix was done.
- Information collection occurred from ten major waste facilities located in Bengaluru city through OpenStreetMap. Data was obtained from ten major garbage management points throughout the Bengaluru city through OpenStreetMap. OpenStreetMap was used to collect data at ten waste management points while also entering garbage values and utilizing a prioritization system of high medium and low values. The research team conducted division of garbage values into three categories consisting of high priority medium priority and low priority.]
- A total of six routing algorithms were employed: Nearest Neighbor as well as Dijkstra's, Bellman-Ford, Johnson's and Floyd-Warshall joined by Ant Colony Optimization. Dijkstra's, Bellman-Ford, Johnson's, Floyd-Warshall, and A route optimization system is provided through the use of Ant Colony Optimization with limits of the capacity of vehicles and the traffic condition of the specific area.
- Time and space were used as measurement factors to determine this aspect of the study. The evaluation considered the elements of space complexity alongside distance traveled and fuel consumption and C02 manufacturing. This information worked towards decreasing systematic waste management sensitivity. The study measured systematic waste management sensitivity through emissions and transportation counts. waste management sensitivity.

This paper has a structure consisting of six separate parts. The second section discusses existing work towards waste collection route optimization as a topic that focuses on methods such as linear programming, GIS implementations, and meta-heuristic methods such as Ant Colony Optimization. Such approaches are oriented for streamlining routes and reducing costs of carrying out routes. Section 3 explains the main system configuration concentrating on the processes of spatial data mapping, garbage load allocation and distance matrix development as parts of data processing for optimising the production of the best routes. In section 4 we focus in detail on the approach and application of six routing algorithms. Nearest Neighbor, Dijkstra's algorithm, Bellman-Ford, Johnson's algorithm, Floyd-Warshall, Ant Colony Optimization. Section 5 discusses system performance by distance, fuel efficiency, CO output, operations time, number of trips and, confirms the fact that the system performs better with reduced costs to environment. Presently, the last part of the study provides conclusions and describes perspectives for further development such as the use of IoT sensors and machine learning for better real-time data management and anticipatory waste management.

II. LITERATURE SURVEY

Christofides et al. [1] Proposed a TSP heuristic, a triangular-inequality-based method which uses spanning trees and perfect matchings, based on Garey and Johnson's work, to be able to improve the worst-case bounds and for efficient polynomial-time approximations.

Auger and Anne [2] provided nature-inspired optimisation techniques for TSP, detailing CMA-ES, genetic programming, and multi-objective evolutionary algorithm to help in overcoming combinatorial complexity, where emphasis was laid on scalability benchmarking through COCO.

Sun and Chutian [3] evaluated metaheuristics for TSP using GA, SA, and ACO as well as hybrid methods of GA with greedy/nearest-neighbor methods and PSO-SA combinations toward improvement of optimization efficiency.

Neman et al. [4] presented a hybrid MIP and time-oriented nearest neighbor model for waste collection optimization in Makkah with 55.7percent enhances in the computation efficiency by using Gurobi Solver and Google OR-Tools.

Algethami et al. [5] designed MTVRP waste collection models that included the aspect of fuel and emissions, by applying the relevant strategies such as CAPSO for tackling the NP-hard optimization issues using the adaptive techniques such as PSO, GA, and Tabu search to the MTVRP waste collection models.

X. Shen et al. [6] developed MSW routing systems, combining economic, environmental and social-related aspects, by exploiting MILP, MOSA, and MOIWOA for multi-objective route optimization and emissions decreasing.

Tirkolaee et al. [7] Presented HFPSO and metaheuristic methods for MSW routing optimization that contribute to improving the environmental and operational efficiency due to the reduced travel distances and respective costs.

Kaya and Serkan [8] solved the MSW collection issues based on CVRP, VRPTW, and GVRP models, where GA, PSO, and modified ACO algorithm with load-sensitive pheromone revisions and a neighborhood scan were applied to reduce emissions and operation costs.

Li et al. [9] evaluated 10 hybrid GA setups for TSP, optimising the method for selection, crossover and mutation, as well as comparing with standard methods such as Roulette Wheel and Stochastic Universal Selection.

So¨zen and Nergiz [10] presents GGSC-SSA, a variant of Sparrow Search Algorithm that incorporates greedy initialization, genetic operators, and sine-cosine control over stagnating and enhancing TSP global optimization.

exact and heuristic methods to TSP were reviewed: GA, SA, TS, PSO, ACO, and ABC, with special attention to hybrid techniques such as GA-ACO and SA-PSO for solving NP-hard combinatorial problem.

Gundu¨z et al. [12] studied both classical and nature-inspired approaches and emphasized hybrid techniques such as ACO along with GA, PSO along with SA for better solution quality in the exact as well as heuristic domain.

Brucal et al. [13] investigated GIS-based TSP solvers in urban logistics, determining the performance limitations of desktop applications for instances of the problem with more than ten locations.

Curtin et al. [14] considered the problem of the PTSP variant of the TSP, when local search and savings-based heuristics are applied, and ACO is introduced as a new nature-inspired method for optimizing the expected length of a path.

Bianchi et al. [15] compare GA and nearest neighbor for reverse supply chain TSP optimization, showing the superior scalability and quality of the solution which one can attain from GA for big complex routing tasks.

SrinivasRao et al. [16] examined metaheuristic algorithms of TSP in e-commerce logistics with regard to the evaluation of the implementations in Python in terms of the measurements such as route length, number of cities, execution time, and error rates for suitability of optimization.

Krishnan et al. [17] proposed E-GTSP, TSP variant with clustering property that can be applied in routing and sequencing problem, transformed into ATSP using the Noon-Bean method to handle practical structural complexities.

Helsgaun and Keld [18] approached set optimal performance (SOP), a precedence-constrained TSP variant, with MILP and dynamic programming for structured route optimization, benchmark validated to TSPLIB using Branch-and-Cut.

Salii et al. [19] studied TSP-PC (Sequential Ordering Problem) within constraint-specific frameworks, suggesting the usage of MILP for noncomplex cases and of dynamic programming for constrained ones in spite of an extremely high usage of memory.

Garc a et al. [20] applied Hopfield networks to large-scale TSP, splitting the problem into sub problems that are solvable, which was however made difficult by the convergence and feasibility problem within the high-dimension instances.

Chen et al. [21] presents the hybrid TSP solvers incorporating the Multi-Objective Deep Reinforcement Learning, Discrete Sparrow Search, and Branch and Bound improved GA performance by means of gene exchange and real-time C-N-GA evaluation.

Ferr˜ ao et al. [22] addressed optimization of CVRP for MSW systems under pressures of urban expansion using data mining and knowledge discovery, a call for integration of eco-driving and Google Maps for real time AI-driven routing.

Hashemi-Amiri et al. [23] applied IoT enabled Smart Waste Management under VRP frameworks and multi-objective metaheuristics to increase the efficiency of recycling procedures and lowering the costs and emissions of operations. Velasquez et al. [24] proposed AI and IoT sensor-based waste collection systems based on QPSO-LR and VRP heuristics to lessen fixed scheduling inefficient and environmental degradation on urban waste operations.
Jwad et al. [25] implemented real-time AI-integrated IoT frameworks for adaptable dynamic waste-collection routing; designing for sustainability, automation and management of operational cost, in spatially extending urban settings.

III. METHODOLOGY

Figure 1 explains the system architecture for optimal waste collection routes. The process starts from collection of location data (latitude and longitude) of the waste bins. Having this data the system creates a structured database and computes distances between points taking into account the containers sizes and vehicle capacity. An algorithm then creates efficient routes abiding by the constraints of operation. It calculates each road in terms of distance and fuel economy to determine a cheapest and practical one. Lastly, the system presents the best route to facilitate improved planning and decision making.

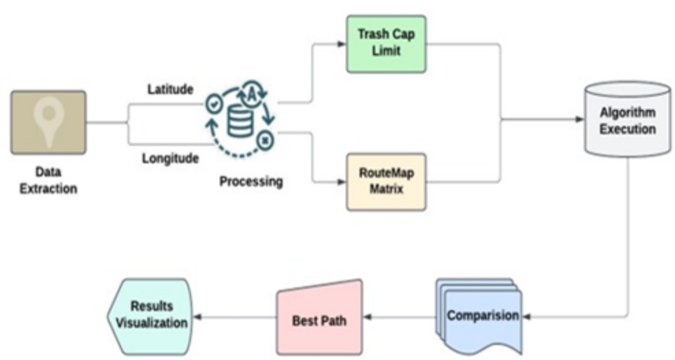

Fig. 1. Fig. 1. System Architecture

A. Data Extraction

The project starts with data collection as its initial step. The acquisition process seeks spatial data about waste collection centers in Bengaluru. Ten major locations serving as data collection areas include Indiranagar and MG Road together with Koramangala and Whitefield along with Electronics City. Jayanagar, Hebbai, Yelahanka, Marathahalli, Basavangudi. By The project implements Google Maps or OpenStreetMap services to obtain geographical data. he data acquisition included the collection of data through OSM that provided coordinates by using latitude and longitude information for these particular locations. The location points were finalised because this approach formed a geographic representation of the waste collection route. waste collection network. A single specific site serves as the place for waste disposal. The conclusion point declared no garbage value for the waste stock. The network features this specific point as the endpoint where waste materials should be transported. This forms the foundation The map for road networks required setup before developing the distance calculations between points. distances between these points for efficient planning.

B. Trash Allocation

The simulation requires data which resembles actual waste generation throughout Bengaluru. A value of 0 kg was assigned as garbage quantity to all collection points. These Notional values were added to the formal database which either came from actual collected data or simulated estimations for the modeling process. A Data Frame structure contains location names together with latitude and longitude information and weight values in kilograms. latitude, longitude, and weight in kilograms of the garbage. The initial value within this model represented zero kilograms of disposed garbage at the disposal point

C. Classification

The waste collection points were grouped into three priority categories according to average waste generation volumes in each of these areas. Locations producing more than 70 kgs of waste were classified as high priority and are colour represented in red. Medium priority, presented in orange, was for those producing 40-70 kilograms. Collection points that had up to 40 kilograms of waste were considered low priority and marked with green. This classification assists in a ranking the routes and does the job of optimizing of waste collection operations properly.

D. Matrix Caluculation

The Haversine formula was used to calculate the distances from all collection points to the disposal land. Distance calculation was inclusive of collection points and disposal point using Haversine formula A mathematical computation determines the separation measured between two geographic locations that considers earth curvature effects. into consideration the curvature of the earth. The computed Analysts used the distance matrix data to determine waste optimization collection routes.

E. Algorithm Execution

A set of six shortest path algorithms produced the best possible method for route derivation best probable routes for waste collection.

Dijkstra's Algorithm: It calculates the shortest path between locations using a greedy approach, which ensures that the vehicle's capacity is not exceeded during the routing process. It's particularly effective for small-scale, simple networks.
Nearest Neighbor: This algorithm finds the proximity to the nearest unknow point and proceeds with this process until all the points are covered or the vehicle is maxed out. It

contributes to smoothing flow by reducing the distance that a vehicle travels and respecting its load capacity.

Bellman-Ford Algorithm: While it works well with graphs that might have negative edge weights, this is unlike Dijkstra's. It is modified to introduce vehicle capacity so that routes can be achieved even under different situations.]

Floyd-Warshall Algorithm: It calculates shortest paths for node pairs for all of the node-pairset and has the ability to work for dense graphs. This method provides a fully optimized run, with the vehicle capacity constraint to identify practical feasible realistic routes for waste collection in the field.

Ant colony optimization: This heuristic algorithm is effective for large-scale dynamic routing problems inspired by the way in which ants discover minimal routes. It adjusts itself to factors such as traffic conditions, time, and a number of vehicles allowing it to be great for optimizing real world waste collection routes.

Johnson's Algorithm: This algorithm identifies shortest path between all nodes and suitable for sparse graph or number of paths is needed. It is especially applicable where there is diversity in traffic density or number of vehicles.

The time complexities of all these algorithms were depicted in TABLE I.

TABLE I
TIME COMPLEXITIES OF ROUTING ALGORTHIMS

Algorithm	Time Complexity
Nearest Neighbor	$O(V^2)$
Dijkstra's Algorithm	$O(V + E \log V)$
Bellman-Ford Algorithm	$O(V \times E)$
Johnson's Algorithm	$O(V^2 \log V + VE)$
Floyd-Warshall Algorithm	$O(V^3)$
Ant colony optimization	Depends on ants and iterations

F. Evaluation Metrics

The algorithms were evaluated based on the following parameters

Execution Time: The total time taken for the algorithm to find the optimal garbage collection route.

Space Complexity: The amount of memory that computer task will take.

Total Distance Covered: The distance traveled by the vehicles added together.

Fuel Consumption: If a car consumes one liter of petrol per 10 kilometers driven the total number of fuel consumption is calculated based on the method

$$\text{Fuel Consumed} = \frac{\text{Total Distance}}{10} \text{ km/litre} \tag{1}$$

CO2 Emission: As a result of application of an emission factor of 2.31 kg of CO consumed per one litre of used petrol, the total CO emissions are calculated as:

$$\text{Total CO}_2 \text{ Emission} = \left(\frac{\text{Total Distance}}{10}\right) \times 2.31 \text{ km/litre} \tag{2}$$

Time complexity:A review of the algorithm's effectiveness, specifically by the ratio of operations to input size.

Number of Trips: The total number of trips needed for the car to collect all the refuse, adhering to its limitation.

IV. RESULTS AND DISCUSSION

Figure 2 is a screenshot of part of the homepage for the Garbage collection route planner- a tool that was developed to make the task of route scheduling easier. The format is very user friendly with options to search with drop down list to choose the start point and a fairly large button titled calculate optimal routes. This tool strives to reduce human participation and enable users to easily create efficient garbage collection routes rapidly. By merely picking the point of origin, the user is provided with optimized routes arising from elaborate computations and therefore easy to the users but very effective in terms of making cities more eco-friendly.

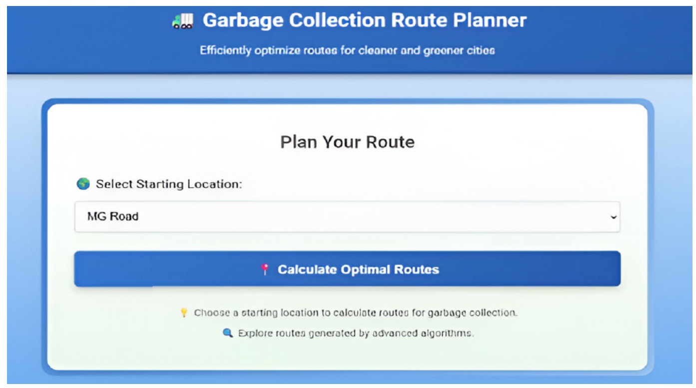

Fig. 2. Fig. 2. Home Page

Figure 3 above illustrates the location selection option of the Garbage Collection Route Planner. Users can select from a list of key localities which include MG Road, Whitefield and Electronic City and have an option to select disposal point. This function allows users to determine the starting point for optimizing routes, which allows users to plan in a flexible way across several areas. Once the location is chosen, users can simply post click and get the best routes, simplifying waste collection.

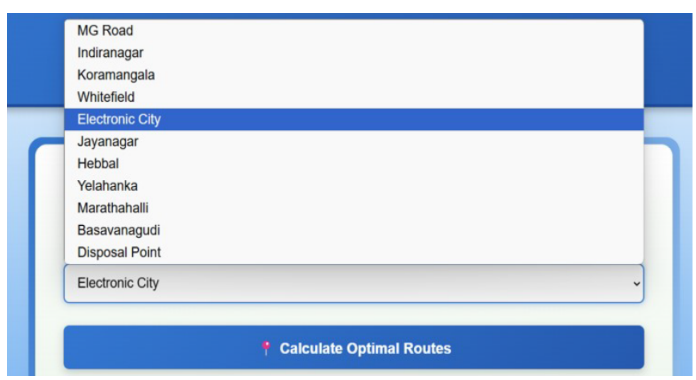

Fig. 3. Fig. 3. Location Selection Feature Page.

Table in Figure 4 shows the performance measures of the Nearest Neighbor Algorithm one of the Garbage Collection Route Planner methods. Such indicators are total distance covered, fuel consumption, CO emission, time taken and number of tours necessary. The attached map demonstrates a path from Electronic City directly to further important locations, such as Koramangala, Jayanagar, and to the disposal point, so creating a path consisting of many collection cycles. The algorithm performs without errors, its time complexity being V^2 and space complexity V, the same performance tables are given for the other five algorithms on the platform. These tables facilitate users' comparison among parameters like route length, fuel consumption, impact on environment, time and efficiency, and accordingly users can choose the best suitable algorithm according to their requirement.

Algorithm	Path Length (km)	Fuel Consumption (L)	CO_2 Emissions (kg)	Execution Time (s)	Number of Trips	Optimal Path	Time Complexity	Space Complexity	Error (if any)
Nearest Neighbor Algorithm	134.64	13.46	31.1	0.0	7	Electronic City -> Koramangala -> Jayanagar -> Basavanagudi -> Disposal Point -> Disposal Point -> MG Road -> Disposal Point -> Indiranagar -> Disposal Point -> Marathahalli -> Disposal Point -> Whitefield -> Disposal Point -> Hebbal -> Disposal Point -> Yelahanka -> Disposal Point	O(V^2)	O(V)	None

Fig. 4. Fig. 4. Evaluation Metrics of algorithm.

Figure 5 presents the map resulting from the Nearest Neighbor algorithm, i.e. the paths created according to the way the algorithm works. Likewise, maps for the other five algorithms will also be presented, but each will demonstrate the special means by which they construct routes.

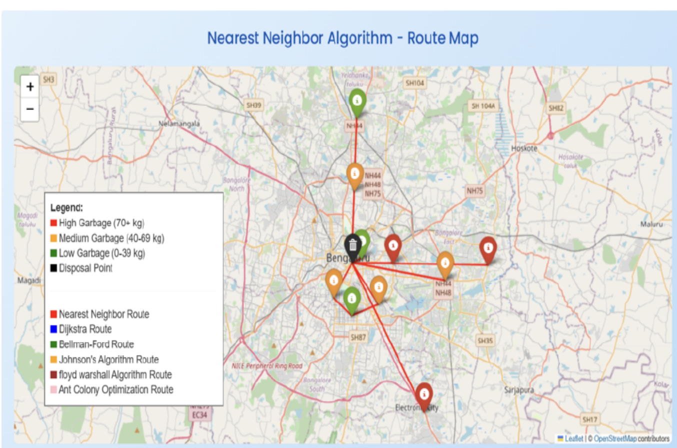

Fig. 5. Fig. 5. Map of Nearest Neighbor.

The Map shown in in Figure 6. A combined map that combines routes generated by all garbage collection algorithms It shows the order of stops according to the selected point of departure and has poivnts of disposal along the best routes. Colorfully marked garbage levels high, medium, or low are indicated by a legend. To clearly represent the individual paths formed by such algorithms like Nearest Neighbor, Dijkstra, and Bellman-Ford variously colored lines are employed. This visual presents dynamics of different algorithms of route planning in easier way, making it simpler for the user to compare functions, and make wiser and more efficient decisions.

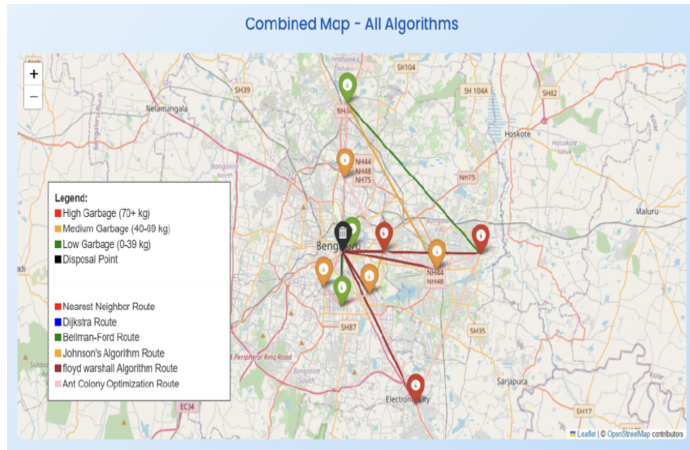

Fig. 6. Fig. 6. Combined Route map for all algorithms.

In Figure 7, we summarize the relevant evaluation metrics, namely, total distance, fuel economy, CO emissions, algorithm duration, number of trips, time complexities, and the space complexities. Additionally, it demonstrates the model of better performance and defines the best algorithm by judging performance from all methods. This enables users to find out which algorithm brings the maximum benefit for garbage collection.

Best Algorithm: Johnson's Algorithm

Path	Total Distance (km)	Fuel Consumption (L)	CO_2 Emissions (kg)	Execution Time (s)	Trips	Time Complexity	Space Complexity
Electronic City -> Koramangala -> Jayanagar -> Basavanagudi -> Disposal Point -> MG Road -> Disposal Point -> Indiranagar -> Marathahalli -> Disposal Point -> Hebbal -> Yelahanka -> Disposal Point -> Whitefield -> Disposal Point	110.48	11.05	25.52	0.0010025501	4	O(VE + V log V)	O(V^2)

Fig. 7. Fig. 7. Evaluation Metrics of Best Algorithm.

V. CONCLUSION

The primary goal was to enhance waste collection efficiency in urban settings through practice environments and fulfilling modern routing algorithms. The selection process was reduced to ten critical points for waste collection with a facility for disposal set aside. The collected data included spatial indicators, such as waste accumulation and distances to other areas with the expectation of improving route optimization. Dijkstra's, Bellman-Ford, Floyd-Warshall, Nearest Neighbor, Johnson's, and Ant Colony Optimization (ACO) algorithms were implemented and analyzed (based on such criteria as runtime, fuel consumption, CO emissions, and overall effectiveness of the journey). We evaluated both the strengths and weaknesses of each algorithm, in association with real life parameters such as road infrastructures, congestion rates, and transport vehicle size. The results provide cities with an opportunity to reduce the costs, as well as optimize delivery schedules and reduce their environmental footprint.

Forthcoming improvements could comprise integration of live data from the IoT sensors on the bins, the traffic control system and the roads, providing dynamic updates of routes based on actual-time waste volumes, traffic and road conditions. The combination of machine learning models might be able to predict waste product and kinds, which might help to plan for waste management strategically. Collaborations with city administrators and the public will improve performance and transparency of urban waste management programs.

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